

WMG Undergraduate 20-Point Student-facing Marking Rubric																				
Student ID: 5602780				Component Title and Weighting: WM279 CW1 – 50%								Marker: Maria Papadaki								

Learning Outcome or Evaluative Criteria	100 %	94%	88%	82%	78%	74%	68%	65%	62%	58%	55%	52%	48%	45%	42%	38%	32%		25%	12%	0%
	Exceptional 1st		High 1st	Upper Mid 1st	Lower Mid 1st	Low 1st	High 2:1	Mid 2:1	Low 2:1	High 2:2	Mid 2:2	Low 2:2	High 3rd	Mid 3rd	Low 3rd	Fail		Low Fail		Fail	
LO2 Apply a relevant risk management approach to a given organisation or scenario	Work of exceptional quality demonstrating outstanding application of risk management frameworks. Appropriate risk management approach is expertly and fully applied in the given scenario, demonstrating exceptional originality, breadth of understanding, and best practice synthesis, with near-publishable insight.		Very high-quality work showing excellent understanding and application of risk management methodologies, with clear, insightful analysis and well-justified best practices across the system.				High-quality work showing clear understanding of methods and generally appropriate best practice choices with some justification, which may lack depth. Risk management considerations are mostly appropriate.			Competent work applying main risk management concepts with limited justification. Some minor errors or inconsistencies. Risk management considerations may be partly appropriate to the given scenario.			Work of limited quality. The risk management approach is partially applied or has notable flaws. There are notable errors or inconsistencies. Risk management considerations are generic or underdeveloped.			Work does not meet the expected standard. Demonstrates some knowledge but with serious omissions in risk management application. There are significant weaknesses or inconsistencies. Risk management considerations are mostly not appropriate.		Work is significantly below standard. Major misunderstandings in risk management, weak or risk management application. Justification is mostly absent or irrelevant		Work is of no academic merit. The risk management approach is missing or completely incorrect.	
LO2 is mapped onto Tasks 2 and 3																					
LO3 Analyse the organisational consequences that result from inadequate information risk management	Work of exceptional quality with flawless analysis of organisational consequences and impacts. The work presents innovative perspectives and critical evaluation supported by relevant examples and seamless structure.		Very high-quality analysis of organisational consequences and impacts with thoughtful evaluation, strong critical thinking and comprehensive use of evidence. Clearly articulated and structured.				High-quality analysis with solid understanding of organisational impacts and consequences. Most impacts are correctly identified with some critical thinking.			Competent analysis that identifies main impacts and consequences. Evaluation and critical analysis may have minor inconsistencies.			Work of limited quality with several errors or omissions. Impacts and consequences may be partially identified. Analysis may be simplistic or inconsistently developed.			Work does not meet the expected standard. It demonstrates basic understanding of information risk management but limited connection to organisational impacts and consequences. Significant omissions and errors present.		Work is significantly below standard. Demonstrates some basic understanding of information risk management but there is no connection to organisational impacts or consequences. The topic lacks clarity, and relevance. Serious omissions and errors present.		Work is of no academic merit. Analysis is either missing, severely flawed, or not submitted.	
LO3 is mapped onto Tasks 1, 2 and 3																					



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What went well? What could be improved and how? *Marker additional comments*

Thank you for your hard work in putting this report together. I have enjoyed reviewing all the submitted items, and identified plenty of positive aspects, as well as areas for improvement in each submission. I have included some general feedback for the entire cohort here, along with details of the marking process, as conducted by the first marker. This is followed by individual feedback and marks for your submission. By the time you review this feedback, the second marking and moderation process will also have been completed, in accordance with the university's marking and assessment policy.

General feedback

All assessments have been reviewed anonymously throughout this process. The initial screening included the calculation of the word count and submission date, as a way of enabling the application of policies on coursework length and late submissions. Subsequently, all submissions were initially reviewed based on their risk assessment methodology, the risk register, identified vulnerabilities/weaknesses, and the top 6 recommendations. No marks were allocated at that initial information gathering stage. Instead, this initial review showed the overwhelming majority stayed within the word limit, adopted an appropriate risk assessment methodology which was not always specified or detailed, and identified relevant risks and recommendations. Very encouraging early indications indeed!

Potential inconsistencies arose from the extent to which the listed vulnerabilities or weaknesses applied to the IIoT devices in scope. In several instances, no vulnerabilities or weaknesses were listed, while in others they did not apply to the manufacturer or the product. In some cases, they were based on incorrect assumptions about the software present in the devices themselves. Although tracking and testing software vulnerabilities and weaknesses in a real-life scenario would be important, I recognise that the only tools available to you would have been the manufacturer advisories and generic software vulnerability repositories, rather than asset inventory tools, software vulnerability scanners or SBOM tools. As such, while individual feedback comments mention potential inconsistencies to that effect, the grading/marketing of LO2 has recognised the difficulty of tracking product vulnerabilities and dependencies.

Another notable area was the extent to which legacy devices had been identified in the risk assessment and addressed through recommendations. Many submissions recognised the risks from managing legacy devices and recommended software/firmware lifecycle management, yet this number could be improved. Additionally, the manufacturer and device security features could also have been considered to a larger extent. For example, manufacturers that have adopted robust firmware/software management practices are likely to have more reported CVEs and vulnerabilities for their products, but also automated signed firmware updates. Similarly, they facilitate tracking of software components and dependencies of their products, which could enhance resilience against software supply chain attacks.

Following the initial review, each individual submission was subsequently reviewed in more detail using the approved marking rubric (see previous page), to assess its alignment with LO2 and LO3, and to identify strengths and areas for improvement. Comments and marks for your individual submission can be found below.

Individual feedback (Student ID: *5602780*)

Marks Breakdown

<i>Details</i>	<i>Weight</i>	<i>Mark</i>	<i>Max</i>
LO2	0.5	68	100
LO3	0.5	68	100
Total	1.0	68	100

What went well

The risk assessment methodology is clearly stated and consistently applied. It is mostly appropriate to the scenario. The adopted approach has been given a great deal of thought and has been linked to threat modelling, threat intelligence and defensive techniques. The review of strengths and challenges of IIoT technologies within warehouses has identified key challenges and best practices within these environments. The heterogeneity and integration of IIoT assets has been recognised as a challenge. The challenge of managing the lifecycle of IIoT software and firmware has been recognised. Misconfigurations in IIoT environments has been recognised as a challenge. Software and firmware patch management has been recognised as a challenge. Asset vulnerabilities have been methodically identified and mapped to D3FEND mitigations. The need for network segmentation has been recognised. The need for zero-trust principles has been recognised. There is excellent use of citations and references.

What could be improved

While the risk assessment methodology has been given a great deal of thought overall, one has to question the assumption that there are no publicly known vulnerabilities for current software and firmware versions of these devices. It then proceeds to model threats, vulnerabilities against assets based on existing knowledge from [11, 6, 7, 8]. The process of doing so could have been described in more detail. Moreover, the implicit assumption that LTS support is available in legacy IIoT devices is incorrect. While the mapping of CVEs to D3fend mitigations is really interesting and creative, it is not clear why ISO27005 has not been used. Is the CVE to D3fend mapping full or partial? Which tool or process was used? Overall, this is a very good submission with excellent features. There is strong evidence of risk assessment best practices, and a creative approach towards mitigating identified risks. Well done.

Student reflections and forward planning. *Space for students to write reflections on their work and note action plans*



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